



Oral Surgery

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(Local Anesthesia)

Lecture 3

“Introduction to Local Anesthesia”

- Many dental procedures, such as tooth extraction, is painful and prolonged and should be performed without pain by using a local anesthesia.
- Local anesthesia has been defined as “loss of sensation in a circumscribed area of the body caused by inhibition of the conduction process in the peripheral nerves”.

Brief history:

- Local anesthesia as it is known today began when a German chemist (Albert Niemann-1860) successfully isolated the active principle of coca leaf, he named it **cocaine**. In 1898 professor Heinrich Braun introduced **procaine** as the first derivative of cocaine which is known as the first synthetic local anesthetic drug. In 1940 the first modern local anesthetic agent was introduced and known as **Lidocaine** (trade name Xylocaine). It relieves pain during the dental surgeries and produces the desired anesthetic effect for several hours.

Terminology:

- **Pain:** According to the World Health Organization (WHO) pain is defined as an “*unpleasant sensation that occurs from imminent tissue damage*”. From a physiological perspective, pain is a warning system. During dental treatment, patients will experience pain as something unpleasant.

- **Analgesia:** is loss of pain sensation unaccompanied by loss of other forms of sensibility.
- **Anesthesia:** loss of all forms of sensation (pain, touch, temperature, and pressure perception).

In dentistry, local anesthesia is used to permit the performance of surgery or other painful procedures with no pain.

- **Paresthesia:** altered sensation (numbness, tingling, burning), such as when a local anesthesia is starting to work or its effect is wearing off. Or when a damaged nerve is regenerated.

- **Pain threshold:-**a point at which the patient feels discomfort when exposed to painful stimuli.
- The patient who feels minimal discomfort from painful stimuli is having a **high pain threshold**, while person with **low pain threshold** is liable to react violently to an identical or even lesser stimulus.



- The pain threshold varies between individuals and in the same individual at different times. This difference in pain threshold is due to several factors such as age, sex and presence of other disease.
- Other factors influence this response such as fear, apprehension and fatigue, all of which lower the pain threshold.

- **Premedication:-** Premedication means giving a drug (or combination of drugs) before a medical or surgical procedure in order to achieve a desired effect. Such as the use of **diazepam** to reduce a patient's apprehension prior to operative treatment.
- **Sedation techniques:** involve the administration of a sedative to reduce anxiety in the conscious patient, usually, given by I.V route.

- The effect of local anesthetics on nerve fiber has been shown to be dependent on:
 - 1- The duration of exposure of local anesthesia
 - 2- Concentration
 - 3- Volume of the solution

Properties (requirements) desirable for ideal local anesthetic:

1. It should not be irritant to the tissue to which it is applied.
2. It should not cause any permanent alteration of nerve structure.
3. Its systemic toxicity should be low.
4. The time of onset of anesthesia should be as short as possible.

Properties (requirements) desirable for ideal local anesthetic:

5. The duration of action must be long enough to permit completion of the procedure.
6. It should be relatively free from producing allergic reactions.
7. It should be stable in solution and should readily undergo biotransformation in the body.
8. It should be sterile.

- Of paramount importance **is systemic toxicity**, because all injectable and most topical local anesthetics are eventually absorbed from their site of administration into the cardiovascular system.

- **Lidocaine**, on the other hand, is effective anesthetic when administered by injection or topical application in clinically acceptable concentrations.
- Several of the more potent injectable local anesthetics (e.g., procaine, mepivacaine) prove to be relatively ineffective when applied topically to mucous membrane.

- **Uses or indications of local anesthesia**

Local anesthesia is indicated in almost all the dental procedures, it is used to render the teeth, supporting tissues and associated structures insensitive to painful stimuli.

A. Diagnostic:

- Administration of LA can be a useful way of finding the source of patient's pain. An example of this is the pain of pulpitis which can be very difficult for both the patient and the dentist to isolate because of its tendency to be referred to other parts of the mouth or face, so LA can differentiate between maxillary and mandibular sources, and even between individual upper teeth provided they are not immediately adjacent.

- Another example is a patient with myofacial pain who is convinced that an upper tooth is causing the problem, LA may help this patient and the surgeon in this situation to eliminate the tooth as the cause of pain and may thus avoid unnecessary treatment.

B. Therapeutic:

- LA can constitute part of a treatment for painful conditions, for example, the use of block technique to eliminate the pain of dry socket. Blocks of the inferior alveolar nerve, mental nerve or infraorbital nerve can also be used for the treatment of trigeminal neuralgia when pain breakthrough.

C. Preoperative:

- The provision of pain-free operation is by far **the most common use** of LA providing an effective and safe method for almost all outpatient dentoalveolar surgical procedures. It can in conjunction with sedation techniques allows more difficult procedures to be carried out without the additional risks of general anesthesia and this is particularly of value in patients with significant cardiovascular or airway diseases.

D. Adjunct to General Anesthesia:

- LA is also given to patients undergoing oral and maxillofacial surgery under general anesthesia this serves the following purposes:
 1. It reduces the depth of general anesthesia needed.
 2. It provides local hemostasis to the operative site.
 3. Provides immediate postoperative analgesia.

Contraindications of local anesthesia

These can be divided into two groups:

1- Absolute contraindications:

- **A. History of allergy to local anesthetic agent**
- Local anesthetic agents belong to the same chemical group should not be used. However, LA agents in the different chemical group can be used. For example, if the patient gives a history of allergy to an ester local anesthetic agent, an amide LA agent should be used.

- **B. History of allergy to other constituents of the local anesthetic solution**
- When the patient has a history of allergy to any of the constituents of the LA solution, it should be avoided, a different type of solution should be used.

2- Relative contraindications:

- A. Uncooperative patient such as mentally retarded patient.
- B. Presence of acute inflammation or suppurative infection at the site of injection, to avoid the spread of infection (with the passages of the needle) from the abscess area to the deeper tissue.
- C. Patient with a significant medical disease such as cardiovascular disease, hepatic dysfunction, renal dysfunction, clinical hyperthyroidism, etc.

Advantages of local anesthesia

1. Less expensive than general anesthesia (GA).
2. No special preparation of the patient is needed as in GA.
3. No complicated apparatus is needed.
4. Less complication rate than GA.
5. The patient remains awake with no need for aftercare.
6. Can be used if GA is contraindicated (the patient is unfit for GA).
7. Anesthetist and other additional trained personnel are not required.
8. The technique is not difficult to master.